

Analyzing Arrogant Comments in the Bengali Language Through Explainable AI and NLP

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Abstract—Detecting arrogance in language is challenging, especially in low-resource languages such as Bangla, where it is often expressed subtly and without overt toxicity. This paper introduces the first large-scale Bangla arrogance detection framework by combining weak supervision with deep learning and explainable AI. The weak supervision framework automatically annotates 46,000 Bangla comments, reducing the need for costly manual annotation. We compare a BiLSTM baseline and the transformer-based BanglaBERT model. BanglaBERT achieved 96% accuracy and a 95% F1-score, outperforming BiLSTM (89.78% accuracy, 87% F1-score). LIME-based explainability revealed that self-referential pronouns and possessive words strongly influenced the arrogance classification. The results demonstrate that our approach effectively detects arrogance in Bangla text and can be adapted for use in other low-resource languages.

Index Terms—Arrogance detection, low-resource languages, Weak supervision, BiLSTM, BanglaBERT, Explainable AI.

I. INTRODUCTION

Online communication platforms, especially social media, have become very popular for people to share their opinions. While much user-generated content is constructive, some carries harmful tones such as hate, toxicity, bullying, abuse, or arrogance. Arrogant comments differ from other harmful speech in that they often convey superiority or self-glorification or make others feel unimportant without using explicit hate or offensive language. They may not use direct hate speech, yet they can still damage respectful interactions [1].

Hate speech is usually aggressive, and toxic comments often use direct insults. Arrogance, in contrast, is context-dependent and tone-driven, often misclassified by existing toxicity detection models [2][3]. Detecting arrogance is further complicated by the lack of annotated datasets in low-resource languages. Existing work on Bangla toxicity detection using both traditional machine learning and deep learning does not address arrogance specifically, leading to misclassification or neglect of this tone.

The challenges in this domain are twofold: (1) the absence of labeled Bangla datasets for arrogance detection, and (2) existing NLP frameworks fail to detect tonal nuances such as arrogance, often misclassifying them as general toxicity, or completely ignoring them. Research in this field becomes more challenging due to the complex

nature of arrogance. Additionally, without explainability model predictions remain difficult to trust for real-world moderation. Although explainable AI (XAI) methods like LIME and SHAP have been applied to other NLP tasks, they have rarely been used to detect subtle tones in low-resource languages like Bengali. This study addresses these gaps through the following research questions:

- **RQ1:** How can arrogance be detected in Bangla without a large manually annotated dataset?
- **RQ2:** Can weak supervision effectively label arrogant expressions?
- **RQ3:** How well can deep learning models detect arrogant tones?
- **RQ4:** Can XAI methods (e.g., LIME) provide interpretable justifications for predictions?

To answer these questions, we create a weakly supervised dataset of 46,000 Bengali comments using keyword-based rules and Snorkel labeling functions. We then train and compare a BiLSTM model and a BanglaBERT model for detecting arrogance, applying LIME to interpret the predictions. The result is a binary classifier capable of both detecting arrogance and explaining its reasoning.

This research can be useful in real life by improving online conversations, making it easier to identify harmful comments and support a more respectful environment. This study also builds a foundation for future NLP work in Bengali and other low-resource languages.

The rest of this paper is structured as follows: Section 2 reviews related studies on toxicity, hate speech, and explainable NLP. Section 3 describes methodology and LIME-based explainability. Section 4 presents results and analysis, and Section 5 discusses key findings, limitations, and future research directions.

II. LITERATURE REVIEW

Research on toxicity, hate speech, and offensive language detection has advanced significantly across multiple languages, including Bengali, using machine learning (ML) and deep learning (DL) techniques. But no prior research has specifically addressed the detection of arrogant comments, particularly in Bengali.

Belal et al. [4] introduced a two-stage deep learning framework for classifying Bengali toxic comments. They used a binary LSTM-BERT classifier achieving 89.42% accuracy for toxicity detection, followed by a multi-label

CNN-BiLSTM which reached 78.92% accuracy. LIME was used for explainability.

Akhter et al. [5] proposed *Bengalibullying*, a hybrid framework that achieved high accuracy in both binary (98.57%) and multi-label (98.82%) classification. However, the model did not address subtle language tones. Similarly, Nath et al. [6] used a BiLSTM model and achieved 95.08% accuracy, but without explainability.

Ashraf et al. [7] used five ML classifiers with LIME for transparency. While Logistic Regression achieved the highest accuracy (85.57%), the study was limited to abusive comment detection rather than subtle arrogance detection. Titli and Paul [8] leveraged Bengali BERT models for abusive text classification (70.6% accuracy), indicating the need for more advanced contextual models.

Chakravarthi et al. [9] developed annotated datasets for abusive comments in Tamil and Tamil-English code-mixed language, comparing classical ML with transformer-based models (MURIL). While transformers excelled in binary classification, classical models performed better in fine-grained abusive categories due to class imbalance.

In Indonesian social media toxic comment classification, Nabilah et al. [10] compared transformer models such as Multilingual BERT, IndoBERT, and IndoRoBERTa Small. IndoBERT achieved the highest F1 score (0.8897), highlighting the benefit of language-specific pretraining for low-resource languages. However, no explainability framework was employed.

Halder et al. [11] improved emotion recognition in Bengali texts using MultiBERT and BanglaBERT with back-translation augmentation, resulting in notable gains in accuracy (8% and 6%) and F1-score (13%). Bidve et al. [12] focused on enhancing transparency in sentiment analysis using LIME. They evaluated LR, RF, and Bi-LSTM models. Bi-LSTM outperformed the other models.

Anand et al. [13] proposed a multilingual offensive language detection system (**MOLD_DL**) combining fuzzy-based CNN feature selection with an ensemble of BiLSTM, Naïve Bayes, and SVM classifiers. Although the model achieved high accuracy (98%) and an F1-score (92.5%), it was lacked interpretability, making it a black-box system.

Rodela et al. [14] employed a hybrid LSTM, GRU, and CNN model with fastText embeddings (91.63% accuracy), while Ahmed et al. [15] highlighted the importance of dataset quality and balanced classes for cyberbullying detection. Islam and Alam [16] introduced BangDSA and skipBangla-BERT for sentiment analysis, achieving 95.71% accuracy, with CNN-BiLSTM performing best among the tested classifiers.

The literature shows significant progress in the detection of toxic, abusive, and offensive language. However, the identification of arrogance remains unexplored. This review highlights the gap in tone-aware explainable NLP systems and proposes a novel approach that combines weak supervision, deep neural models, and XAI to identify and interpret arrogant comments in Bengali.

III. METHODOLOGY

This section details the overall methodology followed in this research. Fig. 1 illustrates the complete pipeline of our research methodology. We first create a weakly supervised dataset. Then, we describe the design and training process of our classification models. Finally, we apply LIME to enhance the interpretability of the model's predictions.

A. Dataset Collection and Pre-Processing

The dataset for this study includes Bengali comments collected from Facebook, YouTube, and online news portals between October 2024 and March 2025, using Selenium. It contains 46,562 comments, including 2,600 AI-generated (ChatGPT and Gemini) arrogant comments. All personal identifiers were anonymized.

Initial analysis showed 434 null entries and duplicate records, which were removed along with irrelevant phrases such as "like," "share," and "join group." After cleaning, the dataset had 46,128 unlabeled comments. We removed punctuation, emojis, and extra spaces to enhance its quality. After cleaning, the dataset was shuffled to ensure random distribution for training and evaluation.

In conventional NLP, stopword removal is a common preprocessing step to reduce noise. It was intentionally skipped in this research due to its significant role in expressing arrogant tones. Removing them risks losing nuanced cues that separate arrogant comments from neutral or factual ones. For instance, in the sentence:

"এই কাজ তোমার দ্বারা সম্ভবই না"

(This task is simply not possible for you.)

Removing stopwords such as 'এই' and 'না' converts the sentence to:

"কাজ তোমার দ্বারা সম্ভবই."

(This task is possible for you.)

B. Weak Supervision Framework

Manual annotation of over 46,000 Bengali comments is impractical due to time and resource constraints. We employed a weak supervision framework combining rule-based functions and the Snorkel labeling model to generate labels for arrogant comment classification.

a) *Rule-Based Labeling*: Initially, a rule-based approach was applied using two sets of lexicons: The arrogant keyword list contained terms that express self-glorification, dismissiveness, or condescension (e.g., 'আমি চাইলে', 'অযোগ্য' etc.), while the hate keyword list included explicit abusive or offensive expressions (e.g., 'হারামজাদা', 'শুয়ার' etc.). Comments matching arrogant keywords were labeled as 1, hate keywords as 0, and the rest as unlabeled (−1). This provided a noisy but useful starting point for weak supervision.

b) *Snorkel-Based Label*: To refine the noisy rule-based labels, we defined three labeling functions (LFs) using Snorkel. Snorkel's PandasLFApplier was used to create a label matrix, which was trained by the LabelModel

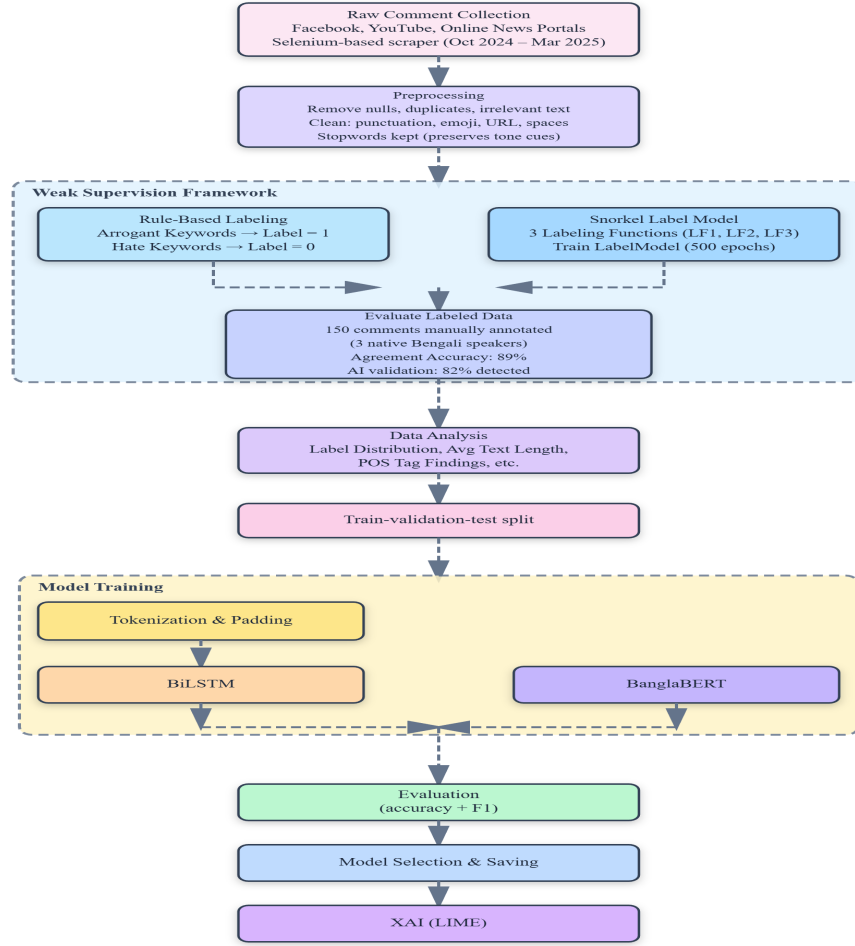


Fig. 1: Model Architecture for Arrogant Comment Detection

with cardinality=3 over 500 epochs to learn LF accuracies and correlations. This refines labels in a new column, `snorkel_label`, improving label confidence with probabilistic outputs. Compared to the rule-based approach, Snorkel reduced ambiguity and improved class balance.

C. Verification of Label Quality

To validate the weakly supervised labels, 550 randomly selected comments were manually annotated by three native Bengali speakers. Fleiss' Kappa = 0.724 indicated substantial agreement, supporting label reliability. Additionally, 82% of 2,000 AI-generated arrogant comments were labeled as arrogant, further confirming consistency.

D. Data Analysis

To understand the structure and distribution of the final labeled dataset, we conducted a comprehensive exploratory data analysis (EDA) on it. The analyses provide insights into the nature of the data.

a) Label Distribution: The final dataset contains 26,096 non-arrogant comments (label 0) and 20,032 arrogant comments (label 1). The class distribution is mod-

erately imbalanced but still allows for effective binary classification using machine learning models.

b) Text Length Analysis: To assess comment complexity, we analyzed word counts for both arrogant and non-arrogant comments. Figure 2 shows that arrogant comments are on average significantly longer than non-arrogant ones. The average length for arrogant comments is 123.81 characters, whereas for non-arrogant comments, it is 70.77. This suggests that arrogant comments tend to be more expressive.

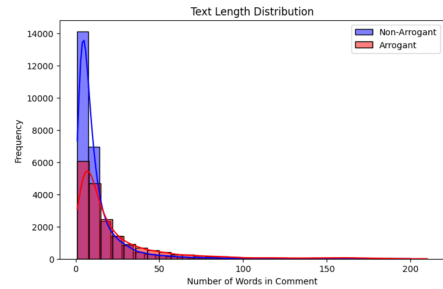


Fig. 2: Histogram of comment lengths by class.

Table I presents a statistical summary of comment lengths. Arrogant comments have a higher mean and standard deviation, indicating greater structural variability.

TABLE I: Statistical Summary of Comment Lengths by Class

Label	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Non-Arrogant	26,096	70.77	99.52	1	21	40	79	1000
Arrogant	20,032	123.81	169.22	3	33	62	137	1767

c) POS Tag and Sentence Complexity: A preliminary POS analysis was performed to understand the linguistic structure of the arrogant comments better. The pilot study revealed some important patterns that these comments often include pronouns such as 'আমি' and 'আমার' indicating self-focus or targeting others, negations like 'না', 'কখনো না', and 'সম্ভব না' enhance dismissiveness, adjectives such as 'সেরা', and 'নিচু' indicate judgment or comparison while verbs such as 'জানি' and 'পারিব না' indicate self-capability.

In terms of sentence complexity, arrogant comments were also observed to have declarative structures such as 'তুমি কিছুই না', conveying direct insult, as well as compound sentences with assertive claims like 'আমি চাইলে পারি, তুই পারিব না', which combine self-praise and dismissal.

E. Model Architecture

This study uses a BiLSTM baseline and the transformer-based BanglaBERT. These were selected to evaluate a traditional deep learning approach with a more advanced transformer method for detecting arrogance in Bangla.

a) BiLSTM-Based Baseline Model: In the BiLSTM model, inputs are first converted into dense vectors using an embedding layer. Then, a two-layer BiLSTM processes the sequence in both forward and backward directions. This helps the model capture meaning from past and future words. A dropout layer with a rate of 0.3 is used to prevent overfitting. The final hidden states from both directions are concatenated and passed through a fully connected dense layer. A sigmoid activation function at the output layer maps the result into a probability score between 0 and 1 (arrogant vs. non-arrogant). Early stopping was applied, and the checkpoint with the lowest validation loss was selected as the best model.

b) Transformer-Based BanglaBERT Model: To capture deeper contextual and linguistic patterns, we employed BanglaBERT, a pre-trained transformer-based language model. It is based on the BERT transformer encoder, which uses self-attention mechanisms to capture deep bidirectional contextual dependencies between words. In this study, we used the “sagorsarker/bangla-bert-base” model from the HuggingFace Transformers library. Text was first tokenized and Each input sequence was padded to a fixed length and passed through the transformer encoder. The contextualized representation of the [CLS] token was then fed into a dropout layer, followed by a fully connected linear layer and a sigmoid activation function for prediction. The model was fine-tuned end-to-end on our annotated dataset.

c) Comparison Rationale: We selected BiLSTM and BanglaBERT to compare the traditional sequential model with the transformer-based architecture. This comparison allows us to evaluate whether advanced language representations (BanglaBERT) offer significant advantages over classical RNN-based models (BiLSTM) in the specific context of Bangla text classification. BiLSTM serves as a lightweight baseline model. It works well for processing sequences and finding patterns over time. However, it struggles with long-range dependencies and deeper context. BanglaBERT, on the other hand, uses self-attention and pre-trained contextual embeddings. This makes it better at handling complex language patterns. This comparison helps us see the trade-offs between simplicity, interpretability, and performance for Bengali language.

F. Training Details

This section outlines the training pipeline, including dataset splitting and necessary training configurations for both BiLSTM and BanglaBERT models.

a) Data Splitting: The dataset of 46,128 comments was divided into three subsets to support model training and evaluation. Approximately 64% (29,521 comments) was used for training, 16% (7,381 comments) for validation during model tuning, and the remaining 20% (9,226 comments) was for model development and evaluation.

b) Hyperparameters: Hyperparameters were chosen carefully to balance performance and generalization. table II below lists the hyperparameters used for both models. Each model was trained using early stopping

TABLE II: Hyperparameter Comparison between Models

Hyperparameter	BiLSTM	BanglaBERT
Embedding Dimension	128	Pre-trained (768)
Hidden Dimension	128	-
Number of Layers	2	12 (inherent to BanglaBERT)
Learning Rate	0.001	2e-5
Optimizer	Adam	AdamW
Patience (early stop)	3	3

based on validation loss or validation accuracy to prevent overfitting.

G. Explainability (Lime)

To ensure the classification results of the model are transparent and interpretable, we employed the Local Interpretable Model-agnostic Explanations (LIME) technique. This is particularly useful for understanding why a given comment is classified as Arrogant or Not Arrogant.

IV. RESULTS AND ANALYSIS

This section reports the experimental outcomes of both the BiLSTM and BanglaBERT models.

A. BiLSTM Performance

The BiLSTM model trained effectively, achieving 93.95% accuracy with a loss of 0.073 during training and 90.09% accuracy with a loss of 0.264 on the validation

set. On the test set, it scored 89.78% accuracy, 91.8% precision, 84.2% recall, and an F1-score of 87%. The advantage of this model is its speed. It takes only about 0.003 seconds to process a comment. This balance of speed and accuracy makes it suitable for real-time use.

B. BanglaBERT Performance

BanglaBERT was able to understand deeper language patterns than the BiLSTM. During training, the model achieved 98% accuracy with a loss of 0.034. On the validation set, the model achieved 96.8% accuracy with a loss of 0.071. On the test set, it scored 96% accuracy, 96% precision, 94% recall, and a 95% F1-score. However, it is slower than BiLSTM, taking 0.163 seconds per comment, and requires more memory and processing power.

C. Comparative Analysis

BanglaBERT outperformed BiLSTM across all metrics, achieving higher accuracy (+4.7%), precision (+5.1%), recall (+4.4%), and F1-score (+4.8%). But this performance gain came at the cost of a 54× slower inference time.

TABLE III: Overall Performance Evaluation

Model	Accuracy	Precision	Recall	F1-Score
BiLSTM	89.78%	91.8%	84.2%	87%
BanglaBERT	96%	96%	94%	95%

While BiLSTM is faster, its ability to model long-range dependencies is limited. BanglaBERT better captures nuances in Bengali comments. The improved performance justifies the higher computational cost of transformer-based models.

D. Error Analysis

To understand model weaknesses, we reviewed some misclassified examples. Both models had trouble with sarcastic or context-dependent comments, where surface-level features were misleading. BiLSTM made more errors on longer sentences, due to its limited memory span. Additionally, class imbalance and noise in the weakly labeled data contributed to some misclassifications. Error Categories and Likely Causes:

- **Sarcastic Arrogance (Missed by BiLSTM):** Indirect or sarcastic arrogance was challenging for BiLSTM. For instance, BiLSTM misclassified the comment 'আপনি তো সবই জানেন, আর কিছু বলার নেই।' as non-arrogant, but BanglaBERT classified it correctly. This happens because transformers can capture long-range context and subtle cues learned during pretraining, which BiLSTM's sequential processing often misses.
- **Short/Slang Comments (Occasional Misclassification by BanglaBERT):** Very short or heavily colloquial comments, such as 'মামা, চুপ কর', which can be ambiguous, were sometimes misclassified by BanglaBERT. This is likely due to limited slang and informal language coverage in the pretraining data.

- **Ambiguous Critical Comments (Disagreement Between Models):** Comments that are critical yet not explicitly arrogant, e.g., 'তুমি কাজটা ঠিক করে নি।' caused disagreement between the models because the tone is context-dependent and borderline.

We also evaluated both models on real-world Bengali comments. Some of them are shown in Table IV below:

TABLE IV: Real-World Comment Predictions

Input Comment	Ground Truth	BiLSTM Prediction	BanglaBERT Prediction
আপনি তো সব জানেন, আর কিছু বলার নেই!	Arrogant	Non-Arrogant	Arrogant
আমি যা চাই, তাই হয়	Arrogant	Arrogant	Arrogant
এই বিষয়ে আমি কিছু বলব না	Non-Arrogant	Arrogant	Non-Arrogant

BiLSTM often misclassifies comments with subtle arrogance due to its limited understanding of context. In contrast, BanglaBERT effectively captures nuanced language patterns, enabling more accurate identification of arrogance in Bengali. These examples highlight the advantage of transformer-based models over traditional sequence models in modeling complex semantics and subtle tones.

E. Explainability Outcomes (LIME)

The LIME analyses revealed insightful patterns in the decision making process of the model. For each comment, it highlighted individual words along with their corresponding contribution scores toward the predicted class.

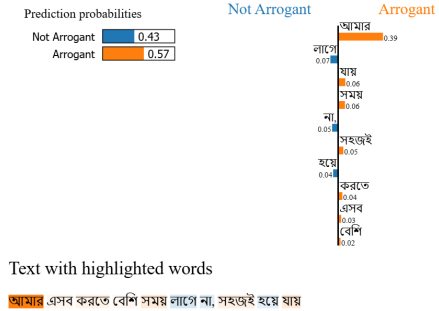


Fig. 3: LIME explanation for an arrogant comment.

For the arrogant example “আমার এসব করতে বেশি সময় লাগে না, সহজেই হয়ে যায়,” the model showed probabilities (57% Arrogant vs. 43% Not Arrogant). The token “আমার” (0.39) had the highest positive influence towards arrogance detection, possibly because first-person possessive pronouns can reflect self-centered or self-promoting tones. On the other hand, “লাগে” (0.07), “না” (0.05), and “হবে” (0.04) contributed small negative weights, softening the arrogance.

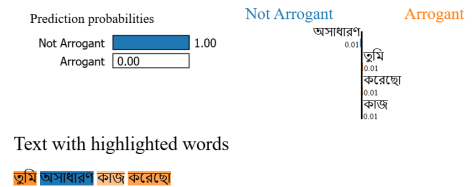


Fig. 4: LIME explanation for a non-arrogant comment.

For the Not Arrogant example “তুমি অসাধারণ কাজ করেছে” the model predicted 100% Not Arrogant. The strongest negative contributors to arrogance detection were “অসাধারণ” (-0.01) carrying appreciative connotations. Interestingly, “তুমি” (0.01) and “ করেছে” (0.01) showed minimal positive weights towards arrogance, possibly because direct address of second-person pronouns.

V. DISCUSSION

This study compared BiLSTM and BanglaBERT for detecting arrogance in Bangla comments using a weakly supervised dataset. Both models were trained on the same dataset splits for a fair comparison. BanglaBERT outperformed BiLSTM across all evaluation metrics. This is due to its transformer-based architecture, which captures longer contextual relationships and understands complex sentence structures and indirect arrogance.

The explainability analysis provided useful insights into how the models make decisions. LIME showed that the model often relies on first-person pronouns, which often signal self-focus and arrogance. However, relying too much on these words sometimes causes errors, especially when pronouns are used in neutral contexts. Error analysis revealed that BiLSTM sometimes misclassifies polite self-references due to limited context, while BanglaBERT confuses sarcasm or irony with arrogance, due to limited representation in training data. Despite these issues, the confusion matrices indicate that both models can generally distinguish between arrogant and non-arrogant comments with few errors.

Despite the strong results, this study has several limitations. First, the dataset was created using weak supervision, which can introduce noisy or incorrect labels. Although the agreement among human annotators was reasonably high, the labels may still be influenced by the rule-based labeling process. Second, the dataset only contains online Bangla comments, which limits generalization to other Bangla text genres. Finally, the explainability analysis mainly focused on individual examples. While LIME helps explain single predictions, it does not fully show how the model behaves overall. Using more advanced explainability methods could help better understand which language features consistently influence the detection of arrogance.

In future, the dataset could be expanded to diverse data. A hybrid model combining BanglaBERT with BiLSTM could achieve higher accuracy. Additionally, tools like attention visualization or SHAP could offer deeper interpretability.

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